THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH3603 Probability theory

Tutorial 5 Solution

1. **Solution.** We are given

$$f(x,y) = \frac{C}{(x+y-1)(x+y)(x+y+1)}, \qquad x, y = 1, 2, 3, \ldots.$$

For an integer $u \geq 2$, there are exactly $u-1$ pairs of positive integers (x, y) with $x + y = u$. Therefore,

$$\mathbb{P}(U = u) = (u-1)\frac{C}{(u-1)u(u+1)} = \frac{C}{u(u+1)}.$$

The total probability must equal 1. Then we get

$$1 = C\sum_{u=2}^{\infty} \frac{1}{u(u+1)} = C\sum_{u=2}^{\infty}\left(\frac{1}{u} - \frac{1}{u+1}\right) = \frac{C}{2}.$$

Thus $C = 2$, and the mass function of U is given by

$$p_U(u) = \begin{cases} \dfrac{2}{u(u+1)}, & u = 2, 3, 4, \ldots, \\ 0, & \text{otherwise.} \end{cases}$$

Since $f(x, y) = f(y, x)$, the distribution of V is symmetric. For $v \leq 0$, we have $\mathbb{P}(X - Y = v) = \mathbb{P}(X - Y = -v) = \mathbb{P}(X - Y = |v|)$. Therefore, we assume $v \geq 0$.

$$\begin{aligned} p_V(v) &= \sum_{j=1}^{\infty} \mathbb{P}(X = j + v, Y = j) \\ &= \sum_{j=1}^{\infty} \frac{2}{(2j+v-1)(2j+v)(2j+v+1)} \\ &= \sum_{j=1}^{\infty} \left[\frac{1}{(2j+v-1)(2j+v)} - \frac{1}{(2j+v)(2j+v+1)}\right] \\ &= \sum_{r=1}^{\infty} \frac{(-1)^{r+1}}{(r+v)(r+v+1)}. \end{aligned}$$

2. **Solution.** Let X and Y be independent with common distribution function F and density function f. The event $\{V \leq t\}$ occurs exactly when both $X \leq t$ and $Y \leq t$. By independence,

$$F_V(t) = \mathbb{P}(X \leq t,\ Y \leq t) = \mathbb{P}(X \leq t)\mathbb{P}(Y \leq t) = F(t)^2.$$

Differentiating gives

$$F_V(t) = F(t)^2, \qquad f_V(t) = 2F(t)f(t).$$

The event $\{U > t\}$ occurs exactly when both $X > t$ and $Y > t$. Hence

$$\begin{aligned} F_U(t) &= 1 - \mathbb{P}(X > t,\ Y > t) \\ &= 1 - \mathbb{P}(X > t)\mathbb{P}(Y > t) \\ &= 1 - \big(1 - F(t)\big)^2. \end{aligned}$$

Differentiating gives

$$F_U(t) = 1 - \big(1 - F(t)\big)^2, \qquad f_U(t) = 2\big(1 - F(t)\big)f(t).$$

The density formulas hold almost everywhere, as usual for densities.

3. **Solution.** We observe that, if $0 < u \le 1$,

$$\begin{aligned} \mathbb{P}(XY \le u) &= \mathbb{P}(XY \le u, Y \le u) + \mathbb{P}(XY \le u, Y > u) \\ &= \mathbb{P}(Y \le u) + \mathbb{P}(X \le u/Y, Y > u) \\ &= u + \int_u^1 \frac{u}{y}\, dy \\ &= u(1 - \log u). \end{aligned}$$

At $u = 0$, the probability is 0, which agrees with the continuous extension of this formula. By the independence of XY and Z,

$$\mathbb{P}(XY \le u, Z^2 \le v) = \mathbb{P}(XY \le u)\mathbb{P}(Z \le \sqrt{v}) = u\sqrt{v}(1 - \log u), \qquad 0 < u, v < 1.$$

Differentiating gives the joint density function

$$g(u, v) = \frac{\log(1/u)}{2\sqrt{v}}, \qquad 0 < u, v < 1,$$

with $g(u, v) = 0$ outside this region.

Hence

$$\mathbb{P}(XY \le Z^2) = \iint_{0 \le u \le v \le 1} \frac{\log(1/u)}{2\sqrt{v}}\, du\, dv = \frac{5}{9}.$$

Arguing more directly,

$$\mathbb{P}(XY \le Z^2) = \iiint_{\substack{0 \le x,y,z \le 1 \\ xy \le z^2}} dx\, dy\, dz = \frac{5}{9}.$$

4. **Solution.** Let $U = X + Y$ and $V = X - Y$. The moment generating function of a $N(\mu, \sigma^2)$ random variable is

$$M_X(a) = M_Y(a) = \exp\left(\mu a + \frac{\sigma^2 a^2}{2}\right).$$

By the independence of X and Y,

$$\begin{aligned} M_{U,V}(s, t) &= \mathbb{E}\left[e^{s(X+Y)+t(X-Y)}\right] \\ &= M_X(s + t)M_Y(s - t) \\ &= \exp\left(2\mu s + \frac{\sigma^2}{2}\big((s + t)^2 + (s - t)^2\big)\right) \\ &= \exp(2\mu s + \sigma^2 s^2)\exp(\sigma^2 t^2). \end{aligned}$$

Setting $t = 0$ and $s = 0$, respectively, gives

$$M_U(s) = \exp(2\mu s + \sigma^2 s^2), \qquad M_V(t) = \exp(\sigma^2 t^2).$$

Hence

$$M_{U,V}(s,t) = M_U(s)M_V(t).$$

Since these moment generating functions are finite in a neighborhood of the origin and uniquely determine the distributions, U and V are independent. Therefore, $X + Y$ and $X - Y$ are independent.